

Object Detection in Dense Volume Data

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Raw Data [1]

- ▶ This project focuses on dense volume data.
- ▶ One of the types of dense volume data is CT scan.
- ▶ CT scans are used in airport baggage screening, assembly lines, and the object detection systems in these places should be able to detect objects fast.
- ▶ Low-resolution CT scanning is fast.
- ▶ Using low-resolution image is one of the ways to address the issue of computational complexity and make the object detection systems fast.

- ▶ Even in the medical domain, to reduce the radiation dose, the exposure time of the patient should be reduced.
- ▶ The exposure time of patients could be reduced by allowing low-resolution CT scans.
- ▶ So, finding out which object detection model has better accuracy at low-resolution CT scans is quite relevant.

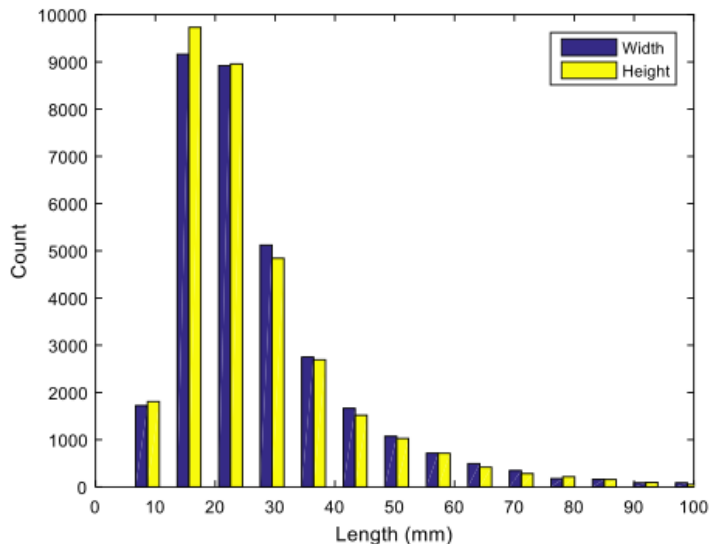
- ▶ Hence in this project, our goal is to analyze the impact of varying resolution of CT scans on both the speed and accuracy of the model.
- ▶ It will also help us to find the model that has the best accuracy at low-resolution CT scans.

Some of the publicly available CT scan datasets are as follows:

- ▶ RSNA pneumonia detection challenge [2]
- ▶ COVID 19 CT scans [3]
- ▶ DeepLesion dataset [1]
- ▶ covid-19-chest-x-ray-lung-bounding-boxes-dataset [4]

- ▶ The dataset chosen should be large enough to train deep learning models.
- ▶ So DeepLesion dataset was chosen as it has 32000 slices of 3D CT scans.

DeepLesion dataset

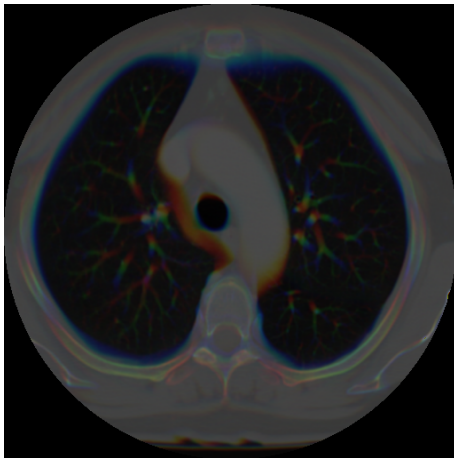


lesion height and width distribution [5]



Raw Data [1]

- ▶ Some of the images were marked as possibly noisy; they were removed.
- ▶ The data processing steps are based on the paper "Improving RetinaNet for CT Lesion Detection with Dense Masks from Weak RECIST Labels"[6] and [7].
- ▶ The images are processed as follows; 32768 is subtracted from each pixel intensity to obtain the Hounsfield Unit (HU) values.
- ▶ HU values are linearly scaled from 0 to 255 and converted into int8 data type.
- ▶ 3 channel image per key slice is generated by concatenating the key slices with a slice 5mm above and below to gain a 3D perspective.



Processed Data [1]

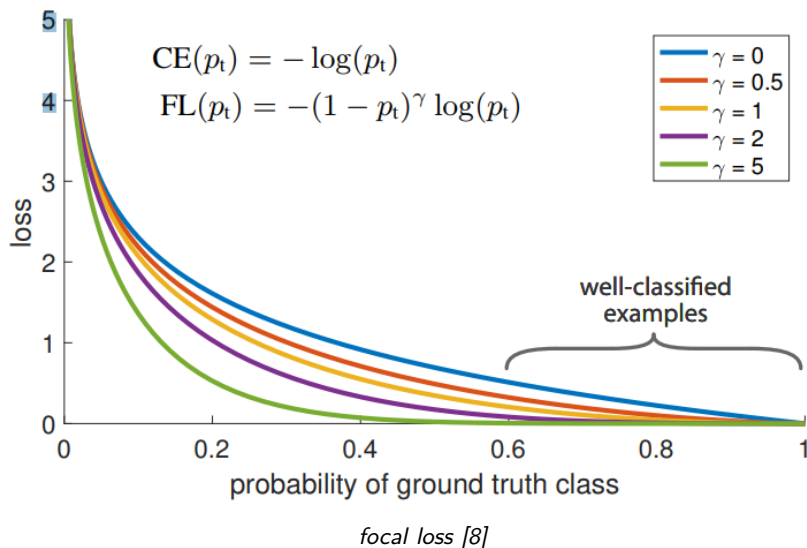
Data distribution	Number of images	Percent(approx)
Train	22901	70
Validation	4889	15
Test	4927	15

Train Validation Test split

We worked on 3 implementations in this project . They are listed below:

- ▶ RetinaNet [8]
- ▶ YOLOv3 [9]
- ▶ YOLOv5 [10]

- ▶ The implementation is based on the paper "Focal Loss for Dense Object Detection" [8].
- ▶ Class imbalance during training is considered the main obstacle impeding one-stage detectors from achieving state-of-the-art accuracy
- ▶ To address the class imbalance, a novel loss function i.e. focal loss function is introduced.



- ▶ RetinaNet utilizes Feature Pyramid Network(FPN) [11] as backbone
- ▶ FPN is placed on top of ResNet [12] architecture to produce multi-scale as well as rich convolutional feature pyramid.
- ▶ RetinaNet consists of two subnetworks.
- ▶ One subnetwork is used for classifying anchor boxes.
- ▶ The other subnetwork is used for bounding box regression.

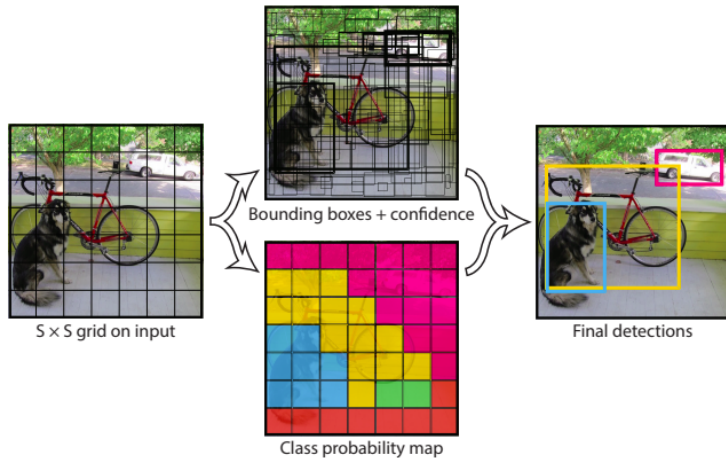
mAP on COCO

	backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
<i>Two-stage methods</i>							
Faster R-CNN+++ [5]	ResNet-101-C4	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w FPN [8]	ResNet-101-FPN	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN by G-RMI [6]	Inception-ResNet-v2 [21]	34.7	55.5	36.7	13.5	38.1	52.0
Faster R-CNN w TDM [20]	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	52.1
<i>One-stage methods</i>							
YOLOv2 [15]	DarkNet-19 [15]	21.6	44.0	19.2	5.0	22.4	35.5
SSD513 [11, 3]	ResNet-101-SSD	31.2	50.4	33.3	10.2	34.5	49.8
DSSD513 [3]	ResNet-101-DSSD	33.2	53.3	35.2	13.0	35.4	51.1
RetinaNet [9]	ResNet-101-FPN	39.1	59.1	42.3	21.8	42.7	50.2
RetinaNet [9]	ResNeXt-101-FPN	40.8	61.1	44.1	24.1	44.2	51.2
YOLOv3 608 × 608	Darknet-53	33.0	57.9	34.4	18.3	35.4	41.9

mAP of various models on COCO [9]

- ▶ Zlocha et al. [6] had shown that RetinaNet gives better results on the DeepLesion dataset than other models like Faster RCNN and Mask RCNN.
- ▶ RetinaNet performs well in the COCO dataset, it has an mAP higher than Faster RCNN on the COCO dataset.
- ▶ So, this model was chosen for this project.

- ▶ Redmon et al. [13] introduced You Only Look Once(YOLO).
- ▶ In the case of YOLO, the task of object detection is taken as a regression problem.
- ▶ The entire process of detection is accomplished using only one network.
- ▶ To achieve this, the entire image is fragmented into $S \times S$ cells.
- ▶ The center of objects is pivotal in the case of YOLO.
- ▶ If the center lies in a particular cell, that cell is accountable for detecting the object.



working of YOLO [13]

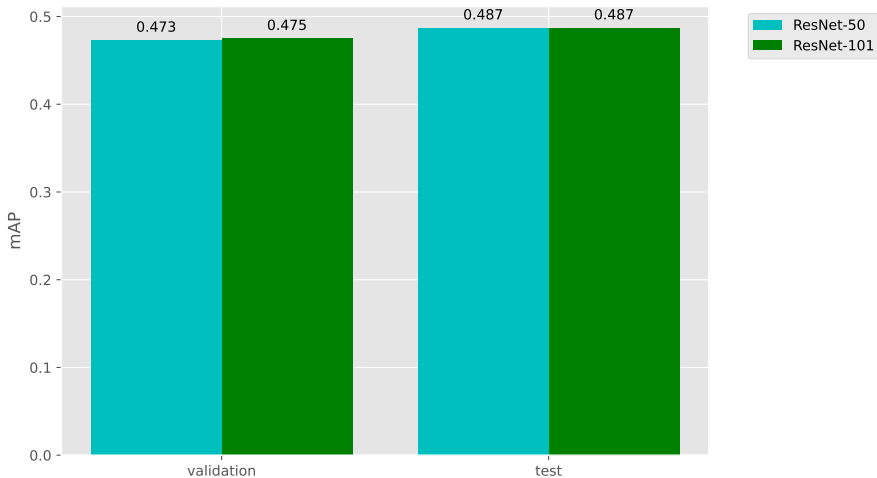
- ▶ The implementation is based on the paper "YOLOv3: An Incremental Improvement"[9].
- ▶ YOLOv3 is an enhanced version of YOLOv2.
- ▶ Some of the improvements that were made on YOLOv2 to get YOLOv3 are listed below:
 - ▶ Logistic regression for confidence scores
 - ▶ Class Prediction
 - ▶ Predictions Across Scales

- ▶ YOLOv3 performs well on the COCO dataset, it has an mAP comparable to Faster RCNN on the COCO dataset.
- ▶ In papers "Automatic detection of pulmonary nodules on CT images with YOLOv3: development and evaluation using simulated and patient data"[14] and "Kidney Recognition in CT Using YOLOv3"[15] YOLOv3 was used for detecting objects in 3D CT scans.
- ▶ Hence this model was chosen in this project.

- ▶ YOLOv5 is an enhanced version of YOLOv3.
- ▶ YOLOv5 was developed by the company Ultralytics.

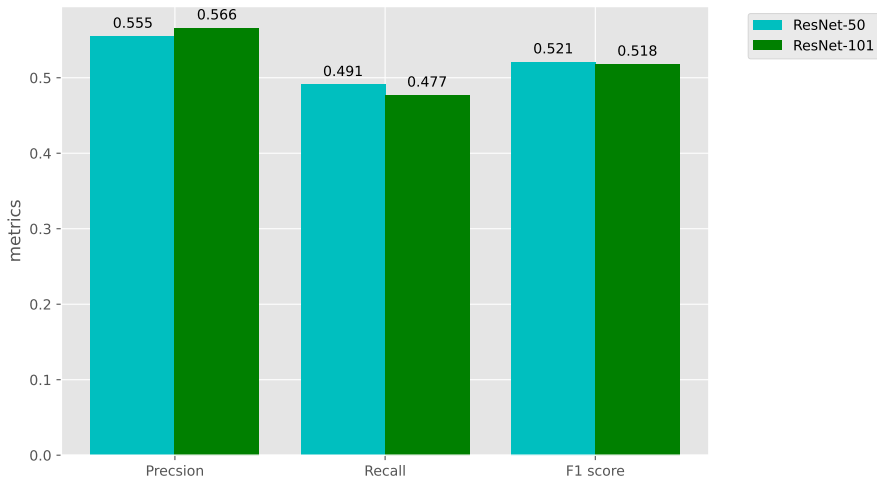
- ▶ K-means clustering
- ▶ Differential Evolution algorithm

Comparison of RetinaNet model with ResNet-50 and ResNet-101 backbone



mAP plot

Comparison of RetinaNet model with ResNet-50 and ResNet-101 backbone

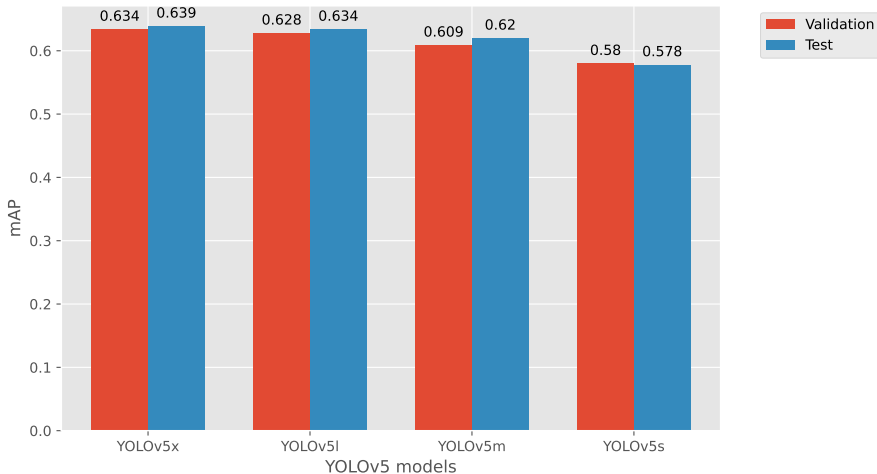


Precision, Recall and F1 score plot

Comparison of RetinaNet model with ResNet-50 and ResNet-101 backbone

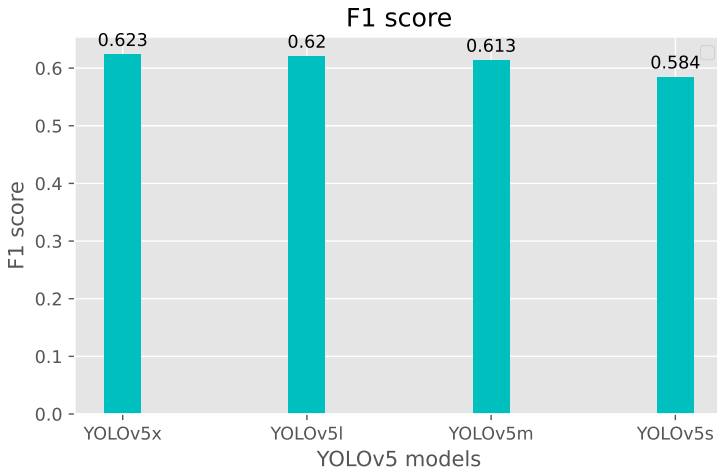
- ▶ ResNet-101 is computationally more expensive than ResNet-50.
- ▶ The inference time of ResNet-50 backbone model on an image from Deeplesion dataset with resolution 512x512 on 64-bit Intel(R) Core(TM) i5-4210M CPU with 2.60GHz frequency is 1.351 seconds and with ResNet-101 backbone is 1.877 second.
- ▶ So, RetinaNet with ResNet-50 backbone was chosen for our experiments.

YOLOv5 models



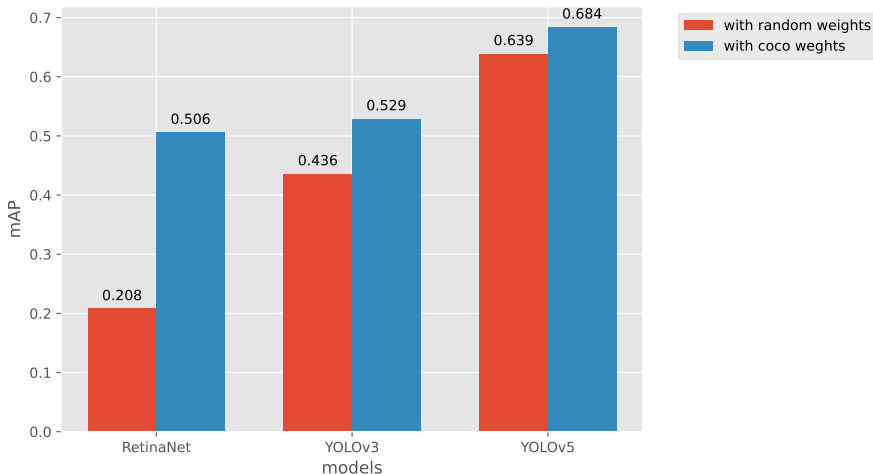
Yolov5 models plot for mAP

YOLOv5 models



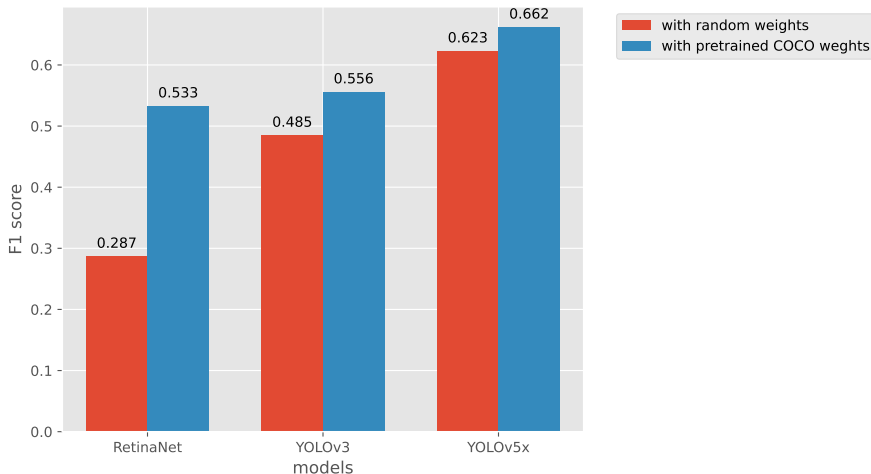
Yolov5 models plot for F1 score

Initialization with random and pretrained weights



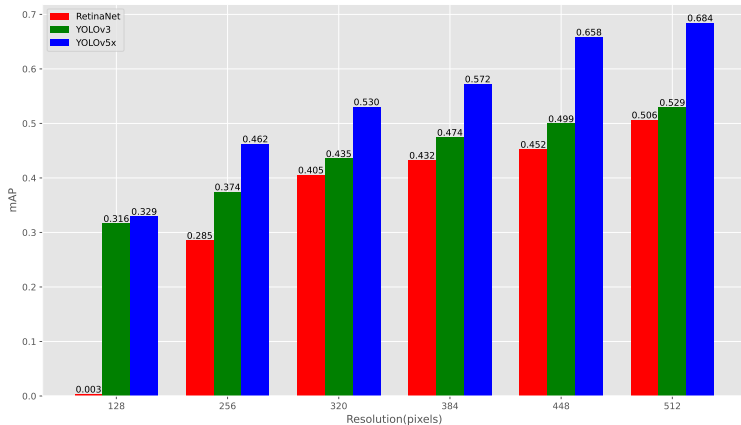
comparison of mAP models with and without initialization weights

Initialization with random and pretrained weights



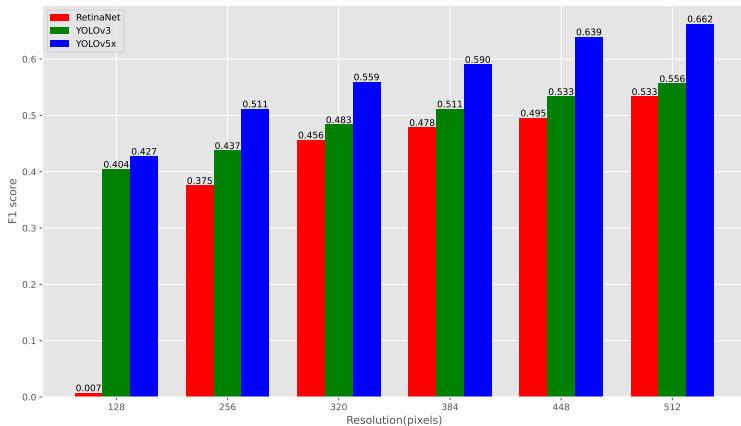
comparison of F1 score of models with random and pretrained COCO initialization weights

Comparison of mAP of models at different resolution



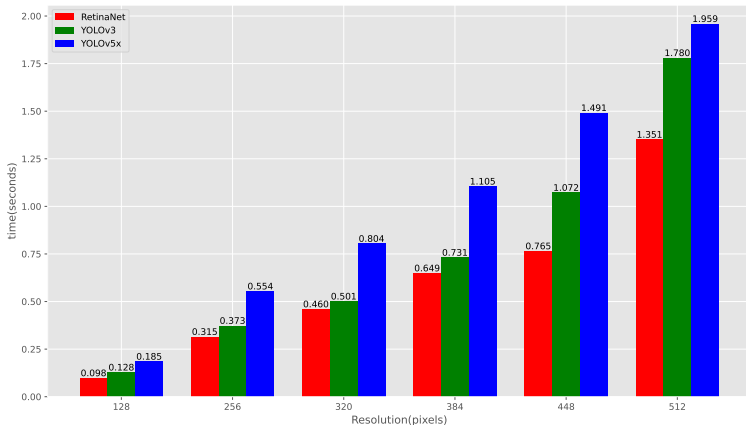
Comparison of models at different resolution

Comparison of F1 score of models at different resolution



Comparison of models at different resolution

Comparison of inference time of models at different resolution



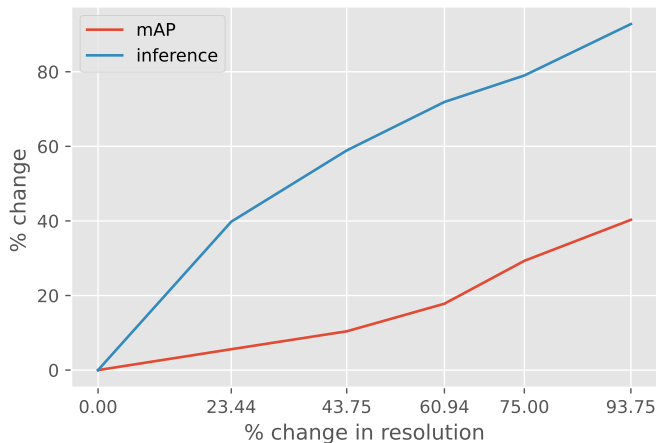
Comparison of models at different resolution

Analysis of Results

Resolution	mAP	Inference time	% change in Resolution	% change in mAP	% change in inference time
512	0.529	1.78	0	0	0
448	0.499	1.072	23.438	5.6	39.8
384	0.474	0.731	43.750	10.4	58.9
320	0.435	0.501	60.938	17.8	71.9
256	0.374	0.373	75.0	29.3	79.0
128	0.316	0.128	93.750	40.3	92.8

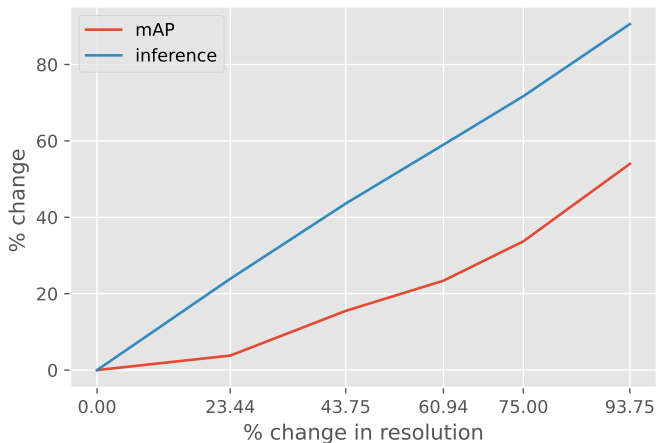
Analysis of Impact of Resolution on inference time and mAP for YOLOv3

Analysis of Results



Analysis of Impact of Resolution on inference time and mAP for YOLOv3

Analysis of Results



Impact of Resolution on inference time and mAP for YOLOv5x

Major Findings

- ▶ YOLOv5x has the best mAP and F1 score at all the different resolutions.
- ▶ RetinaNet is faster than YOLOv3 and YOLOv5.
- ▶ Initializing the model with pretrained COCO weights improved the performance of the model significantly.
- ▶ RetinaNet with ResNet-50 backbone performs better than RetinaNet with backbone ResNet-101.
- ▶ Sacrificing mAP for improving the inference time by reducing resolution is feasible.

- ▶ YOLOv5x has the best mAP and F1 score compared to RetinaNet and YOLOv3 at various resolutions on the DeepLesion dataset.
- ▶ RetinaNet model has the least inference time compared to YOLOv5x and YOLOv3 at various resolutions on the DeepLesion dataset.
- ▶ From the experiments, it could be asserted that sacrificing mAP for improving the inference time by reducing resolution is feasible.

- ▶ Analysis by concatenating multiple slices of volume data
- ▶ Vary resolution with different techniques and analyze its impact on the accuracy and speed of the models.
- ▶ Exploring other ways to make the object detection systems faster.

Thank you very much!

Questions?

References I

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- [7] Aniruddha and P. Saxena. <https://github.com/anir16293/Deep-Lesion>. Accessed on 19 August 2021.
- [8] T.-Y. Lin et al. "Focal loss for dense object detection". In: [Proceedings of the IEEE international conference on computer vision](#). 2017, pp. 2980–2988.
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- [11] T.-Y. Lin et al. "Feature pyramid networks for object detection". In: [Proceedings of the IEEE conference on computer vision and pattern recognition](#). 2017, pp. 2117–2125.

References II

- [12] K. He et al. "Deep Residual Learning for Image Recognition". In: [Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition \(CVPR\)](#). 2016.
- [13] J. Redmon et al. "You only look once: Unified, real-time object detection". In: [Proceedings of the IEEE conference on computer vision and pattern recognition](#). 2016, pp. 779–788.
- [14] C. Liu et al. "Automatic detection of pulmonary nodules on CT images with YOLOv3: development and evaluation using simulated and patient data". In: [Quantitative Imaging in Medicine and Surgery](#) 10.10 (2020), p. 1917.
- [15] A. Lemay. "Kidney recognition in CT using YOLOv3". In: [arXiv preprint arXiv:1910.01268](#) (2019).